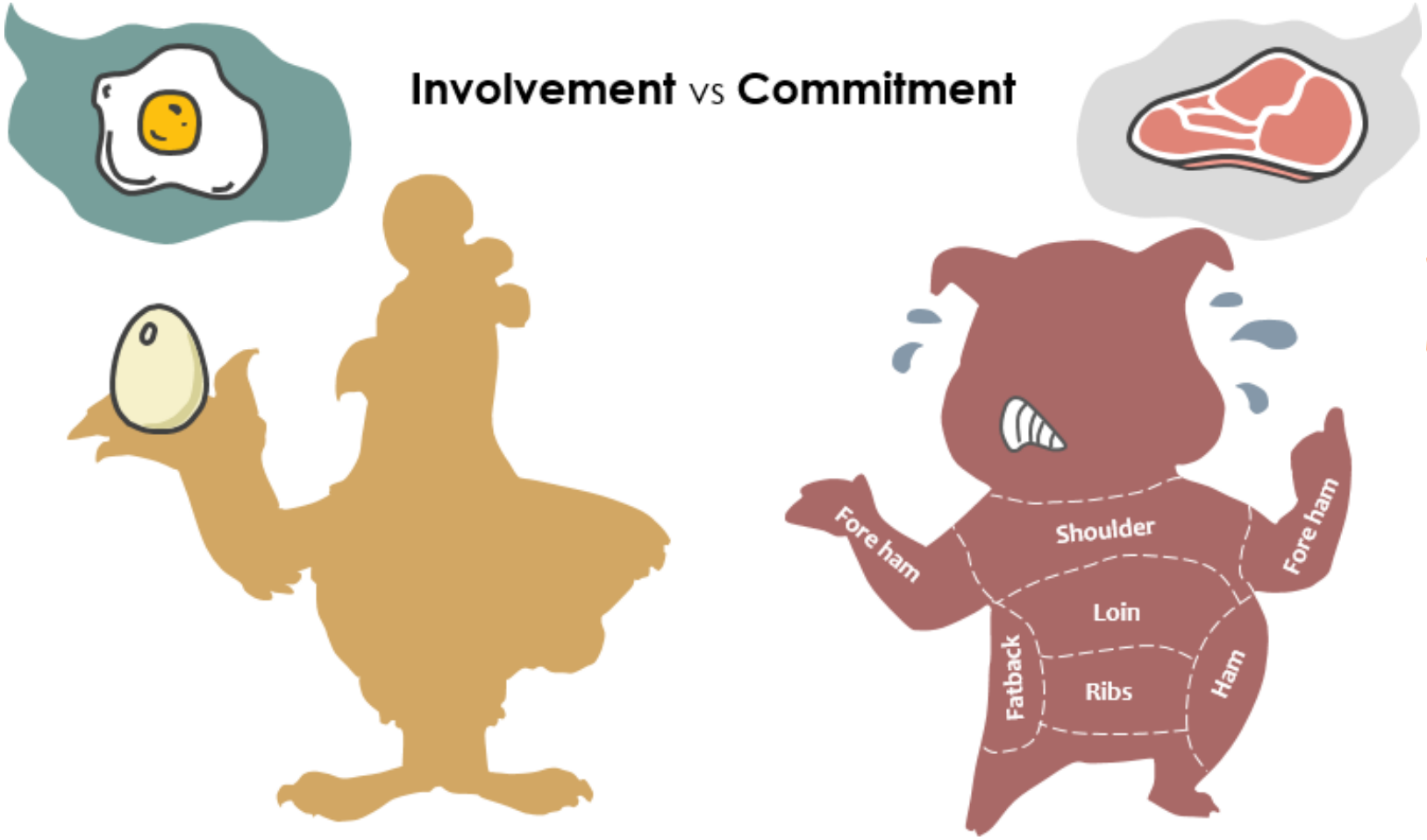


# Advanced Software Development

Johan Bontes (johan@digitsolutions.nl)

CSC3003S




# CSC3003S ASD04 Scrum

A. Intro to Scrum

B. Scrum flow

C. Scrum meetings

D. Process artefacts

E. Scrum risks

# Scrum?

Variant of Agile

Agile: philosophy, Scrum: concrete implementation

State of Agile*: 63% of Agile projects use Scrum [1]

*) just another online survey

[1] http://www.agiledo.cn/assets/img-temp/share/17th-Annual-State-Of-Agile-Report.pdf

# Warning!

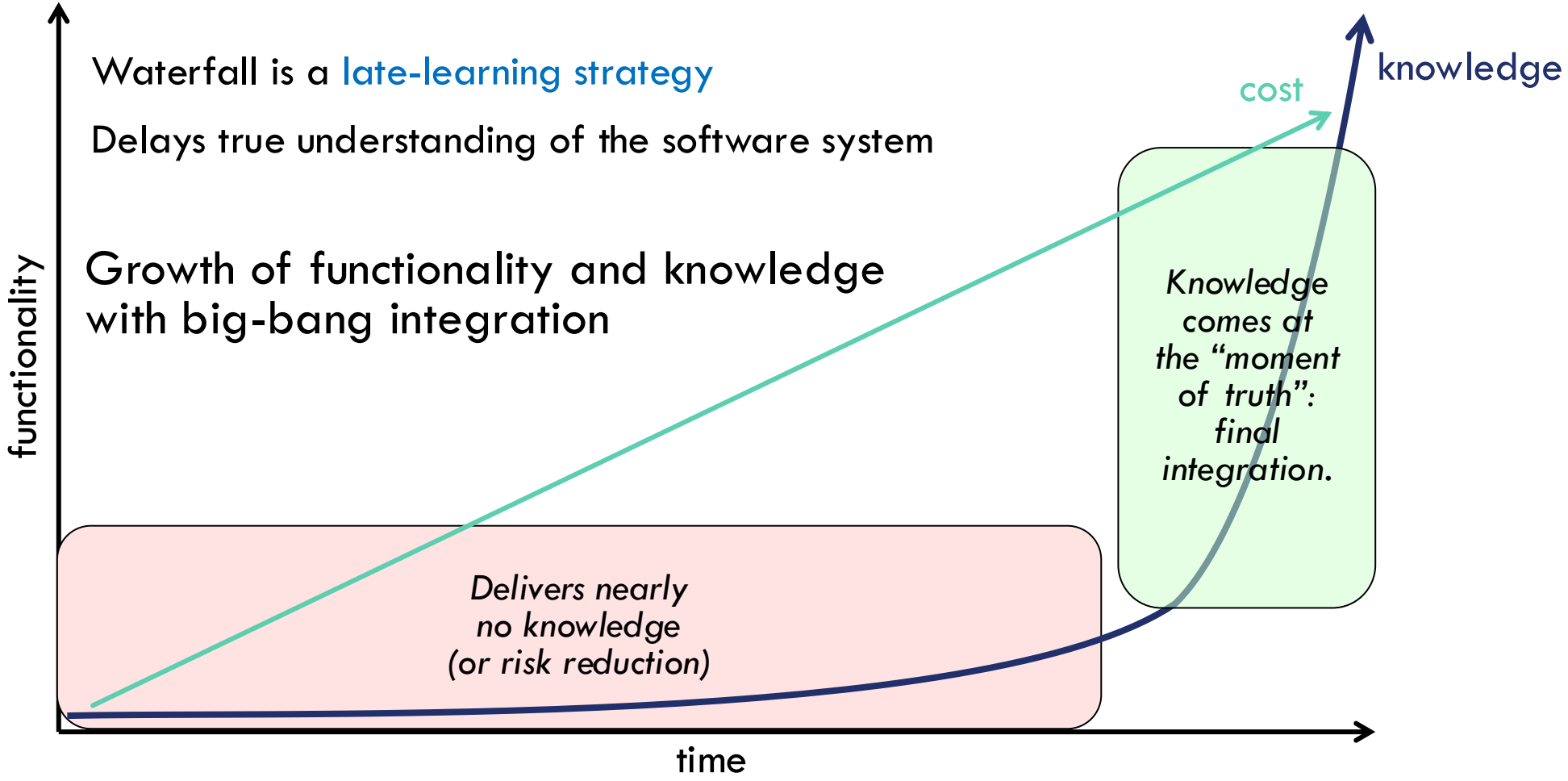
- Not a silver bullet …
  - does not compensate for lack of training or experience
    - Scrum team has defined roles
- Scrum is not a panacea for all known development ills

# Waterfall: Finished functionality

Waterfall is a late-learning strategy

Delays true understanding of the software system

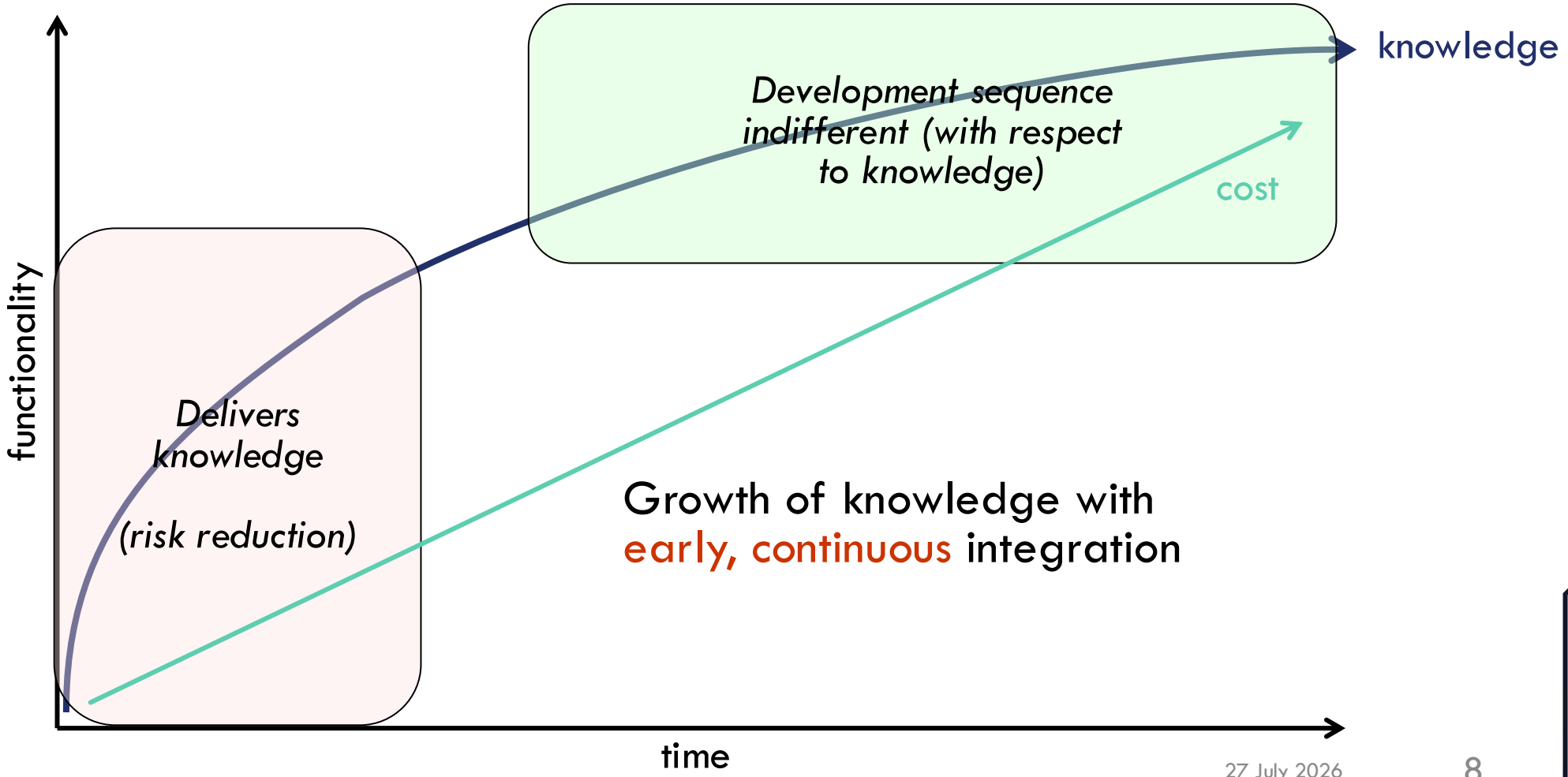
Growth of functionality and knowledge with big-bang integration



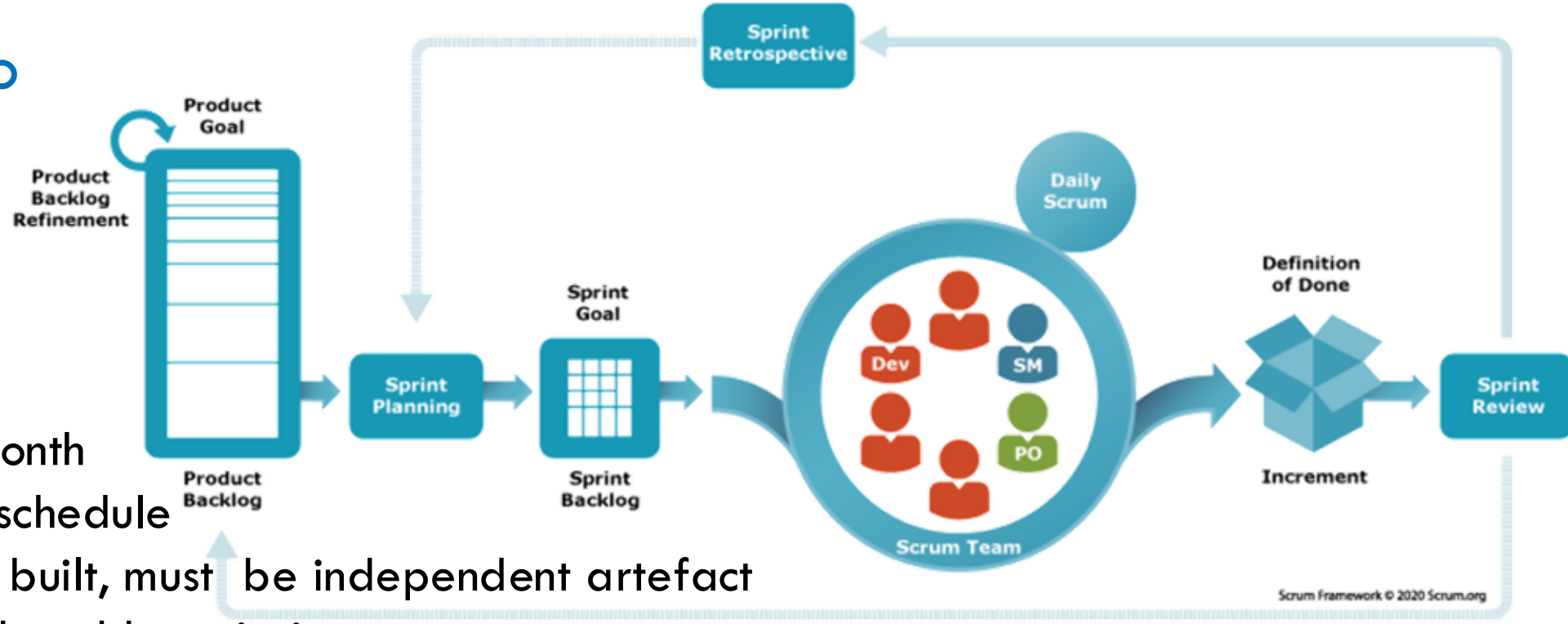
# Scrum: Finished functionality

It pays to learn early in the project

Up front design only reduces risks if problem is well-understood



# Scrum lingo




Goal: end objective

Sprint: fixed time ≤ 1 month

Planning: items to do + schedule

Product: Software to be built, must be independent artefact

Backlog: work items, ordered by priority

Team: ≤ 10 people, no sub teams, no hierarchy

Scrum master (SM): coach and mentor of the team, liaison to management

Daily scrum: 15 min meeting to discuss progress

Increment: outcome of sprint: concrete stepping stone toward the Product Goal

Definition of done: quality as agreed, work finished

Undone work: not finished, quality not OK

Sprint review: Inspect outcome + future adaptations

Sprint introspective: how to improve quality & effectiveness

# Scrum who?

- Team of 10 people or less *(fewer is better)*
  - Need more people? => split product, make multiple Scrum teams

| 1 Scrum master (SM) | 1 Product owner (PO) | *n-2* Developers |
|---|---|---|
| liaison Team ↔ org | can be shared by multiple teams | must have skills needed |
| guards quality | develop project goal | quality work: "definition of done" |
| removes obstacles to progress | plan project backlog + priority | sprint + daily planning |
| chairs scrum events | communicate with customers | training by Pair programming |
| not a leader, but a servant | provides funding | |

# Lingo: Pigs and chickens




- Scrum pigs: Team members (SM, PO, developers)
  - fully committed
- Scrum chickens: customers, users, vendors, management
  - involved: needed, but indirectly
- Scrum is about empowering the pigs and getting the chickens out of the kitchen

# Scrum artifacts?

Artifacts represent either work or value

Each artifact contains a commitment to ensure it provides information that enhances transparency and focus against which progress can be measured:

| | Product backlog | Sprint backlog | Increment |
|---|---|---|---|
| Commitment | Product goal | Sprint goal | Definition of done |
| What is it? | emergent, ordered list of needs to improve product | Set of product backlog items to work on | Concrete stepping stone towards product goal |

| | Product goal | Sprint goal | Definition of done |
|---|---|---|---|
| What is it? | Future state of product for team to plan sprints | Single objective for sprint | Definition of done |
| Characteristics | Clear and concise<br>Transparent in backlog | Flexible in how achieved<br>Makes ≥ 1 increments<br>If impossible: cancel sprint | Formal definition of quality all devs commit to<br>!quality => !done |

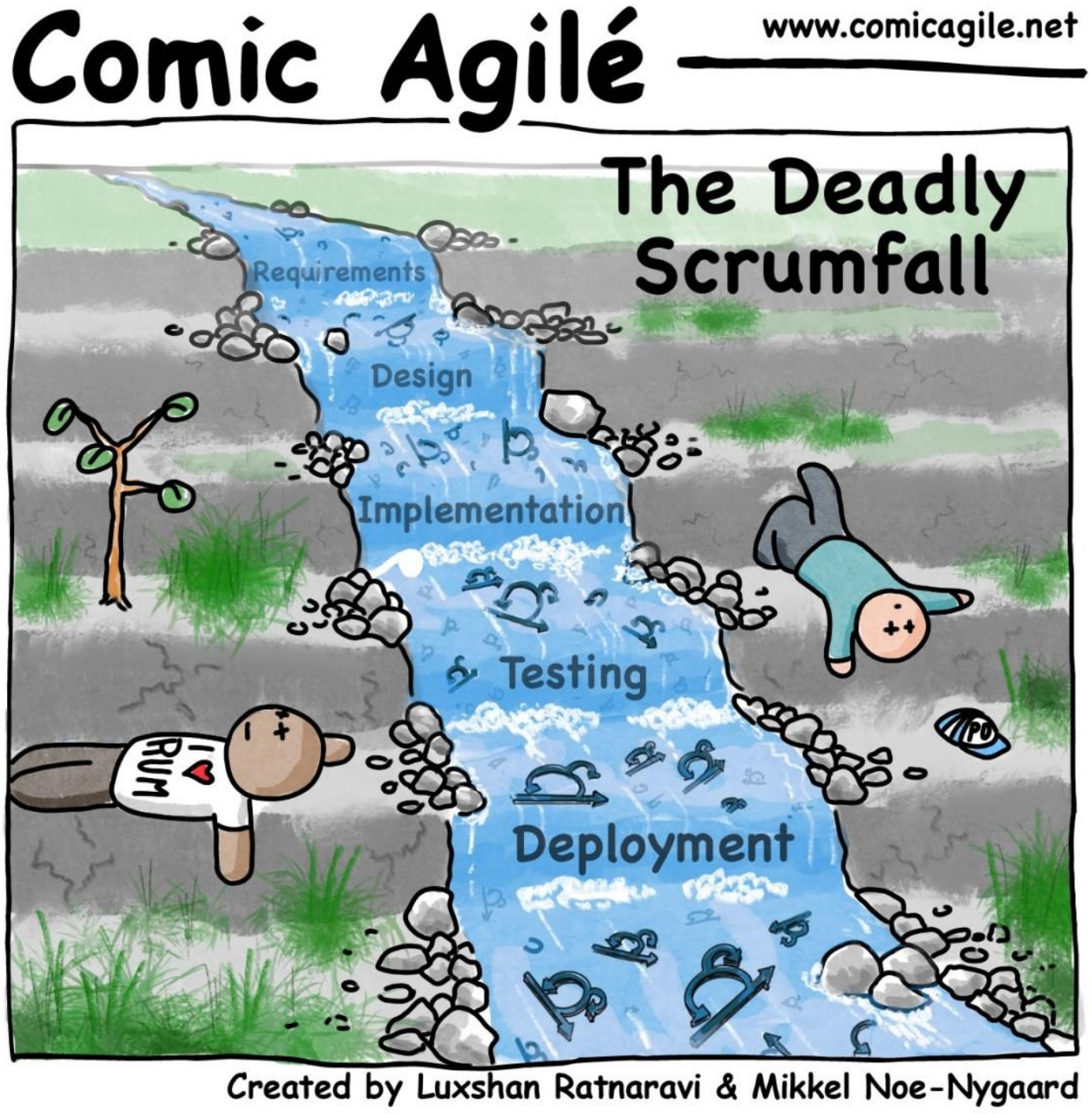
# Scrum Events

- Events only happen inside a Sprint
- Event: formal opportunity to inspect and adapt Scrum artifacts
- All events at regular time and place
- Ideally no meetings outside events

| Event type | Duration | Subject | Output | Participants |
|---|---|---|---|---|
| Sprint | ≤ 1 month (2 weeks) | | Increments | Mostly developers |
| Sprint planning | ≤ 8 hours if 1 month | Product Backlog, Product Goal, Definition of Done | Sprint Backlog, Sprint Goal | Scrum Team |
| Sprint review | ≤ 4 hours if 1 month | Increment, Sprint, Product Backlog, Progress toward Product Goal | Product Backlog | Scrum Team, Stakeholders |
| Sprint introspective | ≤ 3 hours if 1 month | Sprint, Definition of Done | Actionable improvements, Definition of Done | Scrum Team |
| Daily scrum | 15 mins daily | Progress toward Sprint Goal | Progress toward Sprint Goal | Developers |

# Scrum Team

- The team
  - looks at the requirements
  - considers the available technology
  - evaluates its own skills and capabilities
  - determines how to build the functionality, modifying its approach daily
  - figures out what needs to be done and selects the best way to do it.
- The heart of Scrum = *the iterations + creative process*




# CSC3003S ASD04 Scrum

A. Intro to Scrum

B. Scrum flow

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# Scrum Rules

- ScrumMaster = rule-master
- Rules = everyone knows how to play
  - if not enforced: people waste time figuring out what to do
  - If disputed: time is lost while everyone waits for resolution
- Rules work in thousands of successful projects
- To change rules, use Sprint retrospective meeting:
  - Changes should originate from the Team, not management
  - Changes only if the ScrumMaster is convinced
    everyone involved understands how Scrum works.
    - No rule changes until the ScrumMaster
      determines this stage has been reached

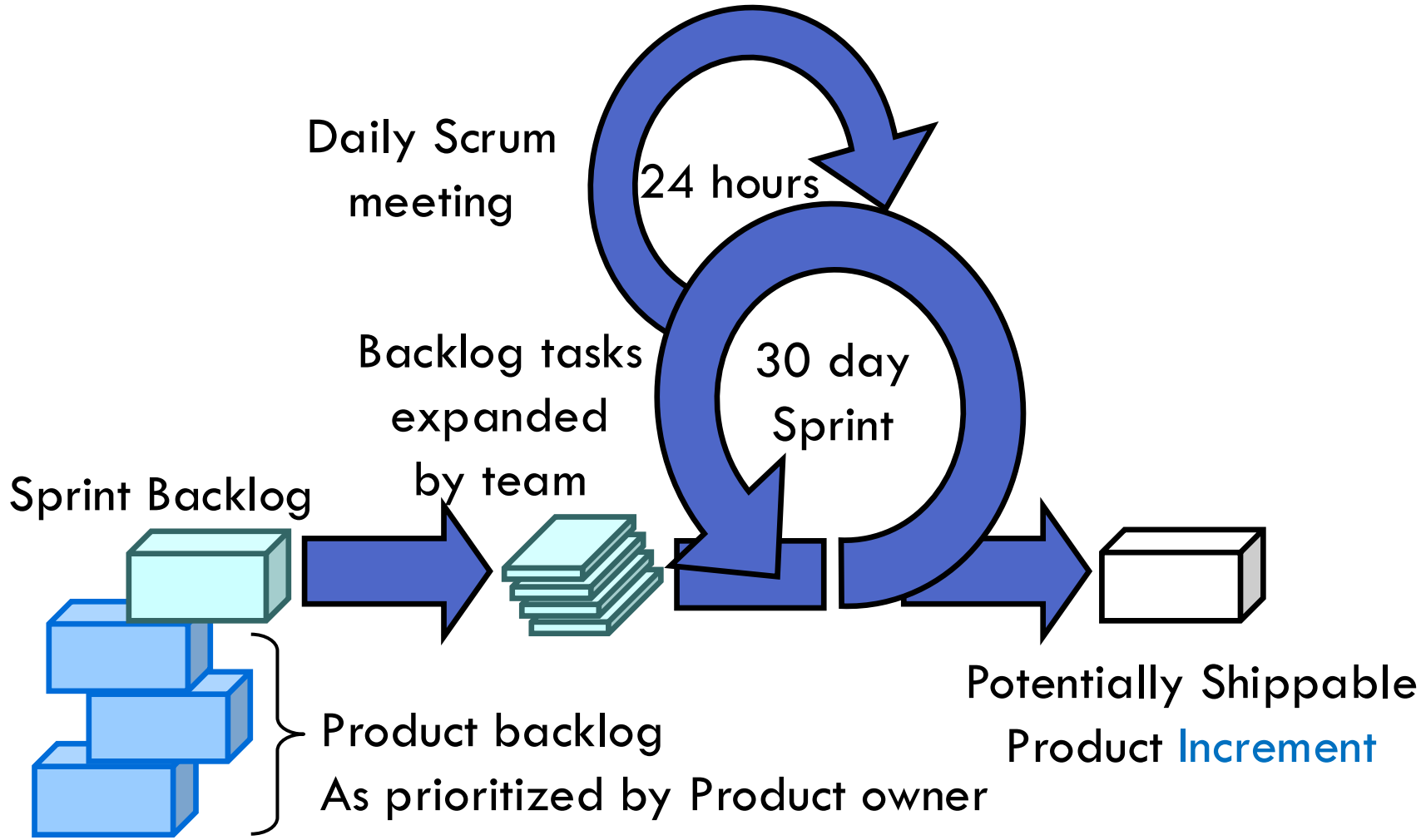
# Scrum Flow

- Any project starts with a vision
  - In XP terms the “System Metaphor”
  - Leads to the logical architecture
- The Product Owner produces the Product Backlog
  - List of functional and non-functional requirements when turned into functionality deliver the vision.
  - Prioritized with the items most likely to generate value as top priority and divided into proposed releases.

# Scrum Flow — Sprints

- Work is done in Sprints.
  - iteration of ≤ 20 consecutive workdays (month).
  - initiated with a Sprint planning meeting
    - the Product Owner and Team collaborate about what will be done for the next Sprint.
      - Select from the highest priority Product Backlog
      - Product Owner tells the Team what is desired,
      - Team tells the Product Owner how much it believes it can turn into functionality over the next Sprint.
    - ≤ 8 hours
      - Prevents hand-wringing about what is possible.
  - Goal is to get to work, not to think about working

# Sprints


Daily Scrum
meeting
24 hours
Backlog tasks
expanded
by team
30 day
Sprint
Sprint Backlog
Potentially Shippable
Product Increment
Product backlog
As prioritized by Product owner



# CSC3003S ASD04 Scrum

A. Intro to Scrum

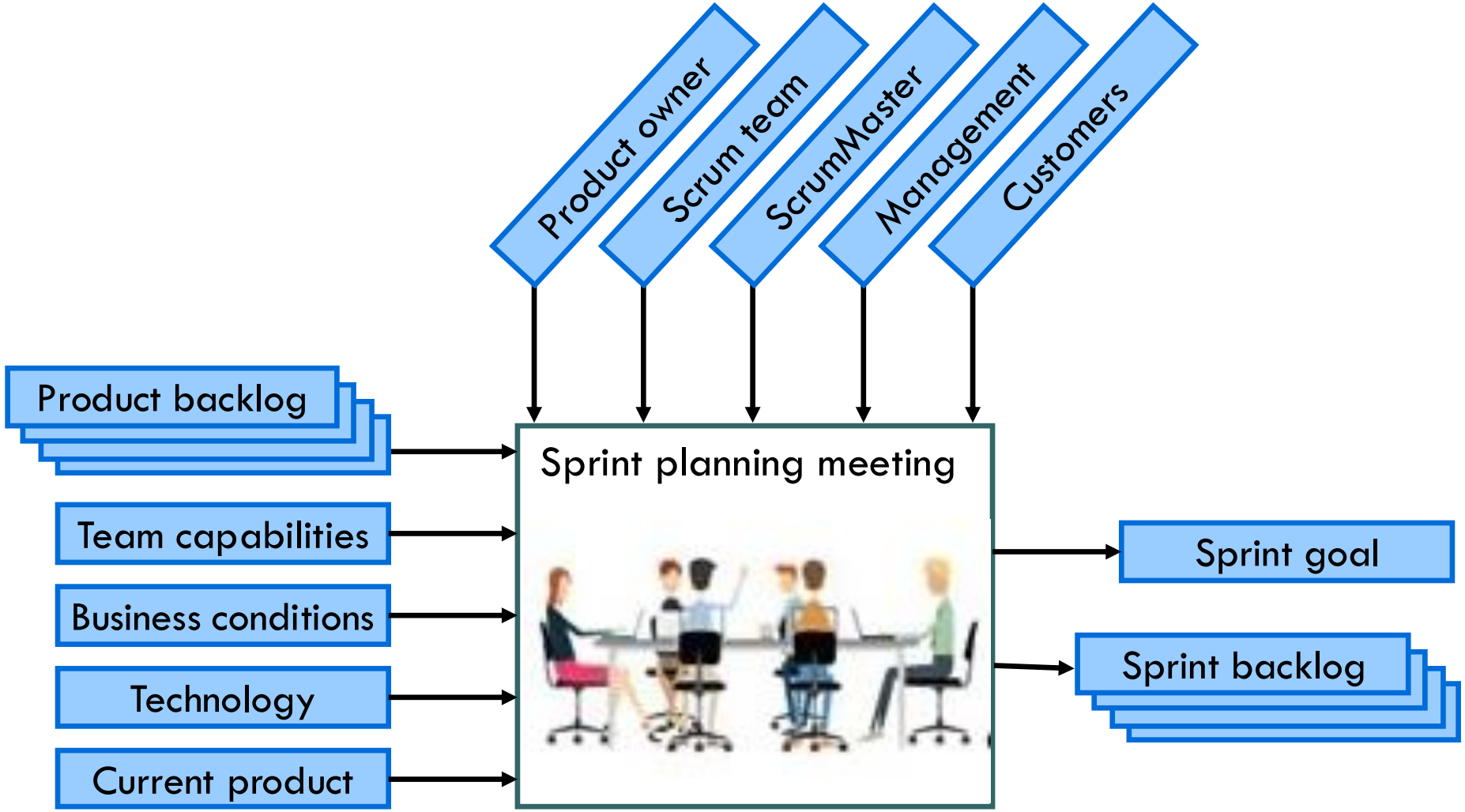
B. Scrum flow

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# Sprint Planning Meeting ≤ 8 hours




2 segments ≤ 4 hours each:

4 hours: select Product backlog;

4 hours: prepare Sprint backlog.

# Sprint planning meeting: 1st segment:

- Select Product Backlog items that it believes it can turn into an increment
- Team will demonstrate this increment to Product Owner and stakeholders at the Sprint review meeting after Sprint.
- Team makes suggestions, Product Backlog selection done by *Product Owner.*
- Team decides how many Product Backlogs to include in Sprint

# Sprint planning meeting: 2nd segment:

- Team manages its own work
- Chosen tasks are placed in a Sprint Backlog
- Product Owner is available to answer questions from the Team
- Team works out how to turn the selected Product Backlog into increment.
- Others only observe or answer questions
- Output of the 2nd segment is the Sprint Backlog
  - tasks, time estimates, and assignments.
- Task list might not be complete
  - it must be complete enough to have commitment of all Team members
  - carries them through the first part of the Sprint,
  - while the Team devises more tasks in the Sprint Backlog.

# Daily Scrum meeting

- Time-boxed to 15 minutes
  - No matter how many team members.
- Held in the same place at the same time every workday.
  - best is first thing in the day, so members think of what they did the day before and what they plan to do today.
- All team members must attend.
  - If a Team member can't attend, must attend by remotely or have another Team member report why
- Starts on time
  - ScrumMaster starts the meeting at the appointed time, regardless of who is present.
  - Any members who are late pay fine to the ScrumMaster immediately.

# Daily Scrum Meeting

- ScrumMaster asks everyone 3 questions
  1. What did you do yesterday?
  2. What will you do today
  3. What stands in the way of getting your work done
- One answerer talk, other people listen
  - Short answers: $\frac{15\ mins}{3\ Qs * ppl\ present} \approx$ 30 seconds per answer
  - no digressions, discussions etc.
- Purpose
  - **Synchronize** work of all Team members daily
  - **Schedule** meetings needed to help progress
- Chickens  can attend, but cannot communicate, nonverbal or otherwise
  - chickens cannot initiate follow-up meetings with Pigs

# Sprint: Time-Boxed

- Sprint is time-boxed ≤ 30 days (more typically 2 weeks)
  - time needed for a Team to build something of significant interest to the Product Owner and stakeholders and bring it to a state where it is potentially shippable.
  - maximum time to do work without artefacts and documentation to support thought processes.
  - maximum time that most stakeholders will wait
    - without losing interest in the Team's progress
    - without losing their belief that the Team is doing something meaningful.

# Sprint

- During the Sprint
  - The Team *can* seek outside advice, help, information, and support.
  - *No unsolicited advice* to the Team.
  - The Team is self-managing.
- The Team has committed to the Product Backlog .
  - The Product Backlog is **frozen** until the end of the Sprint.

# Changing Sprint Backlog

- If Team cannot complete Sprint Backlog
  - can consult with the Product Owner on which items to remove.
    - If too many items removed,
      ScrumMaster can terminate the Sprint

- If Team wants to do more Backlogs
  - Team asks Product Owner which
    Product Backlog items to add.

# Non-Viable Sprint

- If the Sprint proves not to be viable, ScrumMaster may terminate the Sprint and initiate a new Sprint planning meeting.
  - The ScrumMaster can make this change if requested by the Team or Product Owner.

- Sprint can prove to be not viable
  - if the technology proves unworkable,
  - if the business conditions change so Sprint is not of value to the business,
  - if the Team is interfered with during the Sprint by anyone outside the Team.

# Team Responsibilities During a Sprint

- The Team members have two administrative tasks during the Sprint:
  - *attend* the Daily Scrum meeting,
  - keep the **Sprint Backlog up-to-date** and available in a public folder on a public server, visible to all.
    - New tasks must be added to the Sprint Backlog as they are conceived, and the running, day-to-day estimated hours remaining for each task must be kept up-to-date.

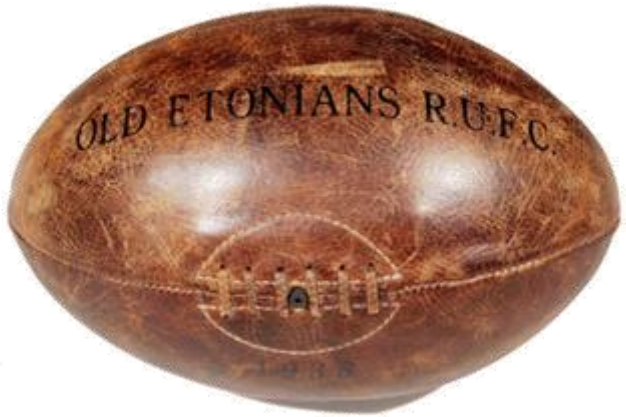
# Sprint Review Meeting ≤ 4 hours

- The Team should spend ≤ 1 hour preparing
- Team presents increment(s) to Product Owner and Stakeholders
- "Definition of Done" means complete & can be shipped or implemented.
  - If “done” has another meaning
    make sure that the Product Owner and stakeholders understand it.

- Anything not 'done' cannot be presented
- Subjects:
  - Sprint goal, Sprint Backlog, Product Backlog completed
  - what went well and what not
  - answer stakeholder questions
  - stakeholder poll to get their impressions, needs, and priority of needs
  - product owner adds and reorders product backlogs

# Sprint Retrospective Meeting ≤ 3 hours

- Attended only by Team, ScrumMaster, and Product Owner (optional)
- All Team members briefly answer 2 questions:
  - What went well during the last Sprint?
  - What could be improved in the next Sprint?
- ScrumMaster summaries the answers
- Then Team prioritizes the order of issues and talks about potential improvements.
- The ScrumMaster facilitates Team's search for improvement of Scrum process
- Actionable items can be added as high-priority non-functional Product Backlog items.




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# Scrum's Artefacts

- Scrum has artefacts
  - Product Backlog
  - Sprint Backlog
  - Burndown Charts
- Can be managed using an Excel spreadsheet or cloud-based tools
  - Consider support for distributed teams

# Product Backlog

- Prioritised list of first cut refinements.
- Product Owner is free to
  - adjust the order in which Product Backlog items are developed
  - add new items
- This brings in the concept of agility and change

| Backlog item | Estimate (person-weeks) |
|---|---|
| **Break props into separate rigid bodies** | **1** |
| **Arrow projectiles stick to environment** | **0.5** |
| **Ragdolls can lose limbs** | **2** |
| **NPCs avoid dynamite thrown at them** | **4** |
| **Improve exception handling** | **0.5** |

# Sprint Backlog

- List of the things that will be “done” during the Sprint
- Each item has an estimate of how long it should take to complete, usually measured in hours.
- During the Sprint’s 30 day period, the Project Team must update the Sprint Backlog
- Keeping the Sprint Backlog updated is key:
  - not only does it allow us to work out how fast a team can work (their velocity), it is an early warning indicator.

# Sprint Backlog Update

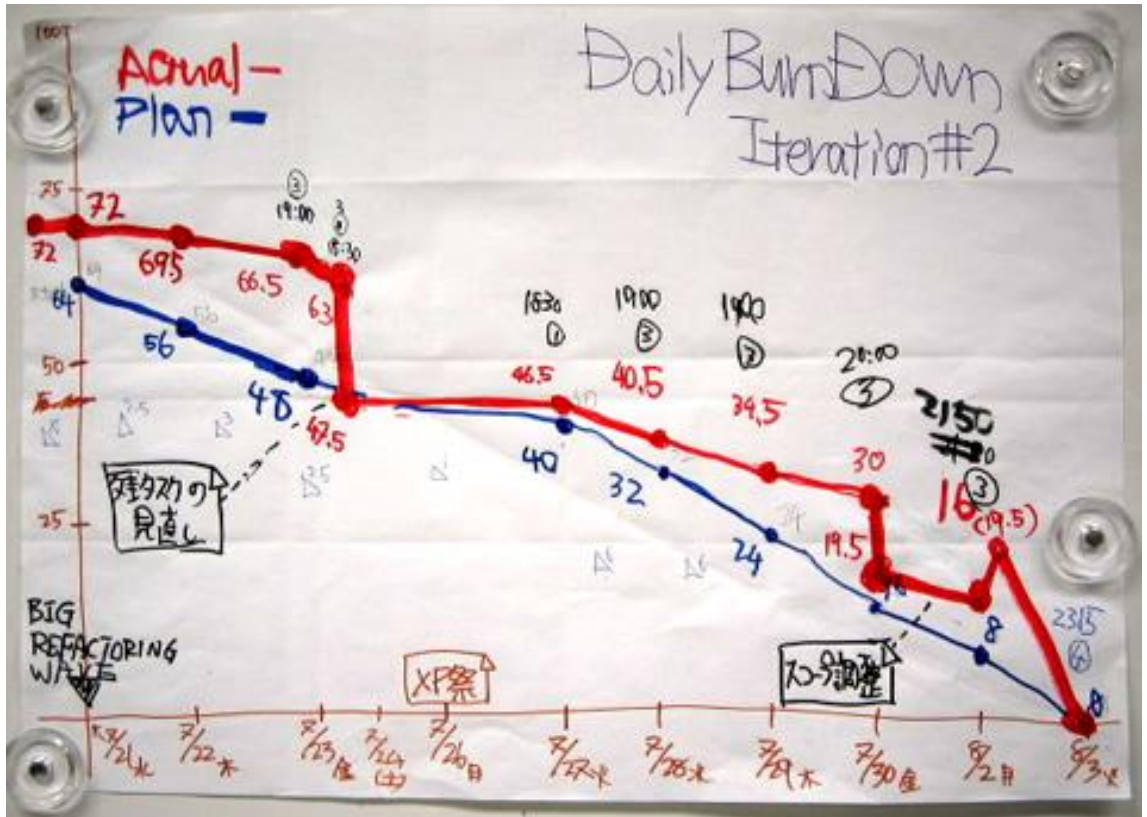
| Sprint 1 | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | |
| 9/8/2006 | | | | Sprint Day | 1 | 2 | 3 | 4 | 5 | | | 6 | 7 | 8 | 9 |
| | | | | | Mo | Tu | Wed | Thur | Fri | Sat | Sun | Mon | Tue | Wed | Thur |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| | | | | Hours | 152 | 144 | 136 | 124 | 116 | | | 112 | 106 | 96 | 84 |
| | **19 days work in this sprint** | | | Remaining | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| ID | Backlog Item | Description | Owner | Estimate | | | | | | | | | | | |
| 1 | Minor | Remove bad formatting in login screen | GJ | 8 | 8 | 4 | 2 | 0 | | | | | | | |
| 2 | Minor | Correct legacy code in reorts | KJ | 8 | 8 | 4 | 0 | | | | | | | | |
| 3 | Minor | Modify user file handling | CM | 20 | 20 | 16 | 8 | 4 | 2 | | | 0 | | | |
| 4 | Minor | Add network code for error cases | CM | 40 | 40 | 36 | 32 | 30 | 26 | | | 24 | 20 | 20 | 18 |
| 5 | Minor | Add company logos to all screens | DB | 20 | 18 | 16 | 12 | 12 | 12 | | | 12 | 12 | 12 | 12 |
| 6 | Major | Add algorithm to remove jitter | JG | 56 | 56 | 56 | 56 | 56 | 56 | | | 56 | 56 | 56 | 56 |
| 7 | Major | Change Design for workflow | DC | 72 | 64 | 60 | 60 | 60 | 56 | | | 56 | 50 | 42 | 42 |
| 8 | Super | Supervision/Guidance | CAM | 32 | 32 | 28 | 28 | 24 | 24 | | | 22 | 20 | 20 | 16 |

# Burndown Charts

- These are used to identify the rate that work is being completed.
- *Graphical representation of the sprint backlog*
- It illustrates the work done.
- Demonstrate a steady drive to zero hours remaining:
  - it represents a pace of work that should be sustainable.
  - however, some work takes longer than others,
  - some are even shorter,
  - may not be a perfect straight line.

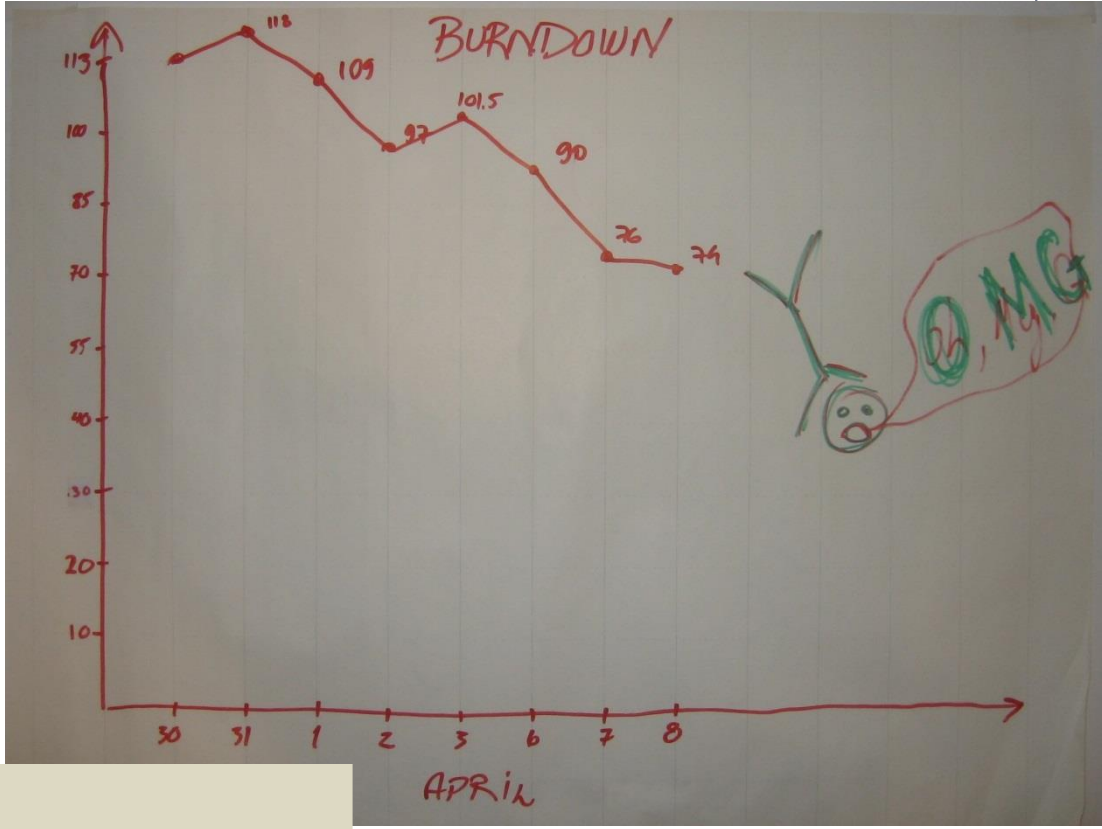
# Burndown Chart

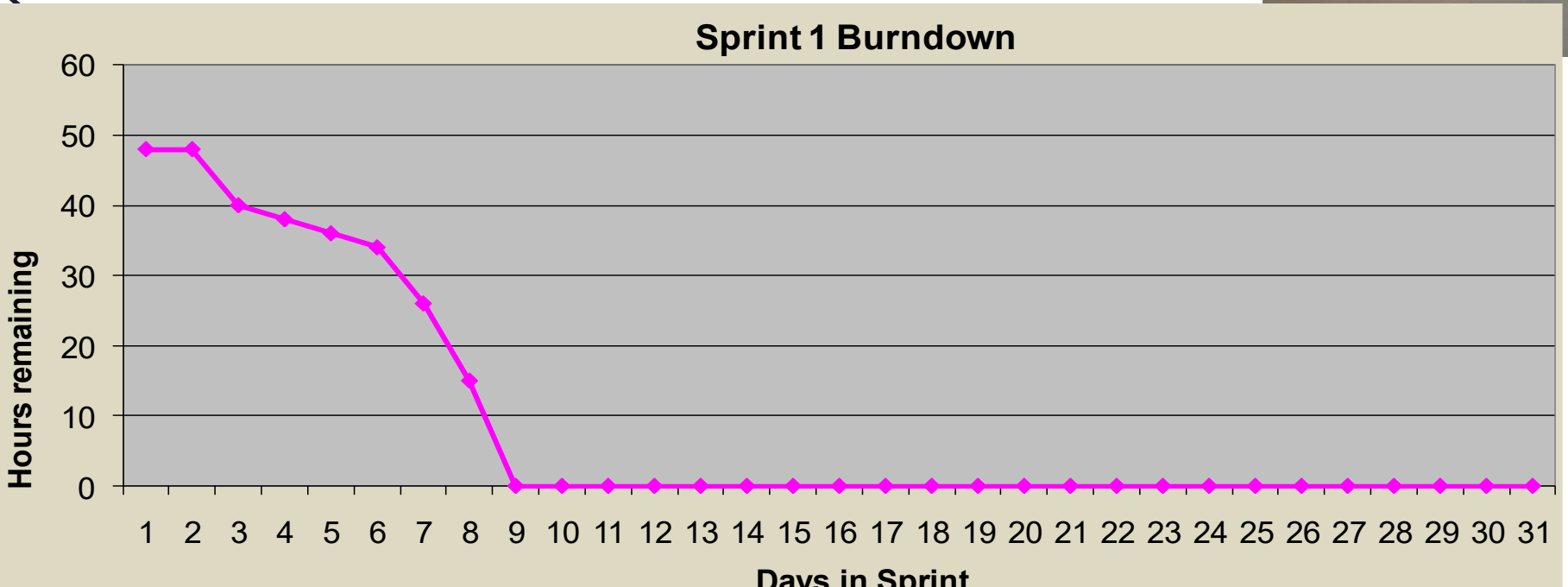
- Initially in a sprint there is no work done so the chart is as follows
- It is easy to see whether the work is being done in time.
- The following indicates it is not being done fast enough



# Burndown Chart – example

- Too slow, due to:
  - inexperienced developers
  - hard roadblocks
  - distractions
  - bad planning




Or to fast




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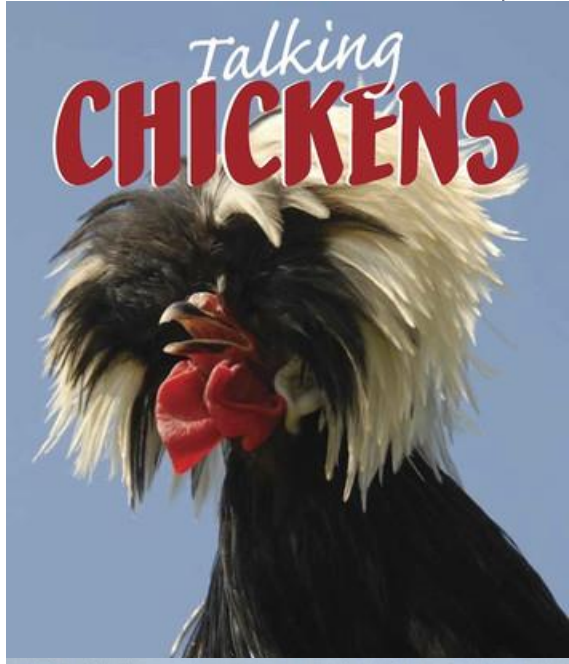
E. Scrum Problems

# What can go wrong: *Loss of Rhythm*

- Sprints are not always the same length.
  - On a well-executed Scrum project the team establishes a natural rhythm.
  - Each sprint must be the same length.
  - Different length Scrums break rhythm
    - Sprints feel like arbitrary units of time selected more by outside forces not designed to improve productivity of the team.
    - Makes it hard for teams to plan

# What can go wrong: Talking Chickens

- Chickens at daily Scrum ask questions or make observations.
  - Only pigs are allowed to speak
  - Talking chickens: slippery slope.
    If a chicken is allowed useful comment,
    how do we later prevent non useful talk?
  - Have to draw the line, best to be strict




# What can go wrong: Missing pigs

- Not all pigs attend the daily Scrum meeting.
  - Some companies have flexible hours
  - Flex time, and the night work habit of developers,
    makes it common to have some members of a team late.



# What can go wrong: *Missing Pigs II*

- it can be difficult for a team meet for a daily Scrum every day.
  - a daily meeting, at the same time in the same place each day help a project establish and maintain its rhythm.
  - If too many pigs miss daily Scrums too often then perhaps the meeting is too long or deviating from the 3 standard questions.
- When run well, daily Scrum ≤ 15 minutes and useful for every pig.
  - There can be specific deadlines that requires people to skip a daily Scrum
    - only in emergencies
  - when too many pigs miss meetings is there a problem

# What can go wrong: *Specialized Job Roles*

- A project team has specialized job roles such as Architect, Designer, DBA, or Tester.
  - Scrum teams need to have a "*we're all in this together attitude*"
    - This can be undermined if a team has specialized job descriptions or roles.
  - It is simplistic to think that everyone can be a DBA and everyone can write server-side J2EE or .Net code.
- A successful Scrum team need not be all generalists.
  - Each specialist must accept responsibility for the system as a whole.

# Wrong: *the Daily Scrum is for the ScrumMaster*

- The Daily Scrum is a status update from team to ScrumMaster.
  - ScrumMaster must be a servant
  - ScrumMaster takes notes about who committed to what work and why some other task wasn't completed.
  - Feels like status meetings
- Two main purposes of the daily Scrum
  - to provide a coordination to all on the project.
    - Everyone knows where everyone else is.
  - Each team member makes commitments in front of his peers.
  - If the commitment is not fulfilled, ScrumMaster's does not rebuke the team member.
    - They should feel bad enough.

# Wrong: *ScrumMaster assigns work*

- Work is assigned by the ScrumMaster

  not signed up for by developers.

  - Self-organization is one of the underlying principles of Scrum.
  - If ScrumMaster assigns work, it undermines the responsibility of developers; they self-organize to achieve the goal.
  - Even occasional assignments by ScrumMaster are harmful
  - Teams must feel completely in control of their own work.